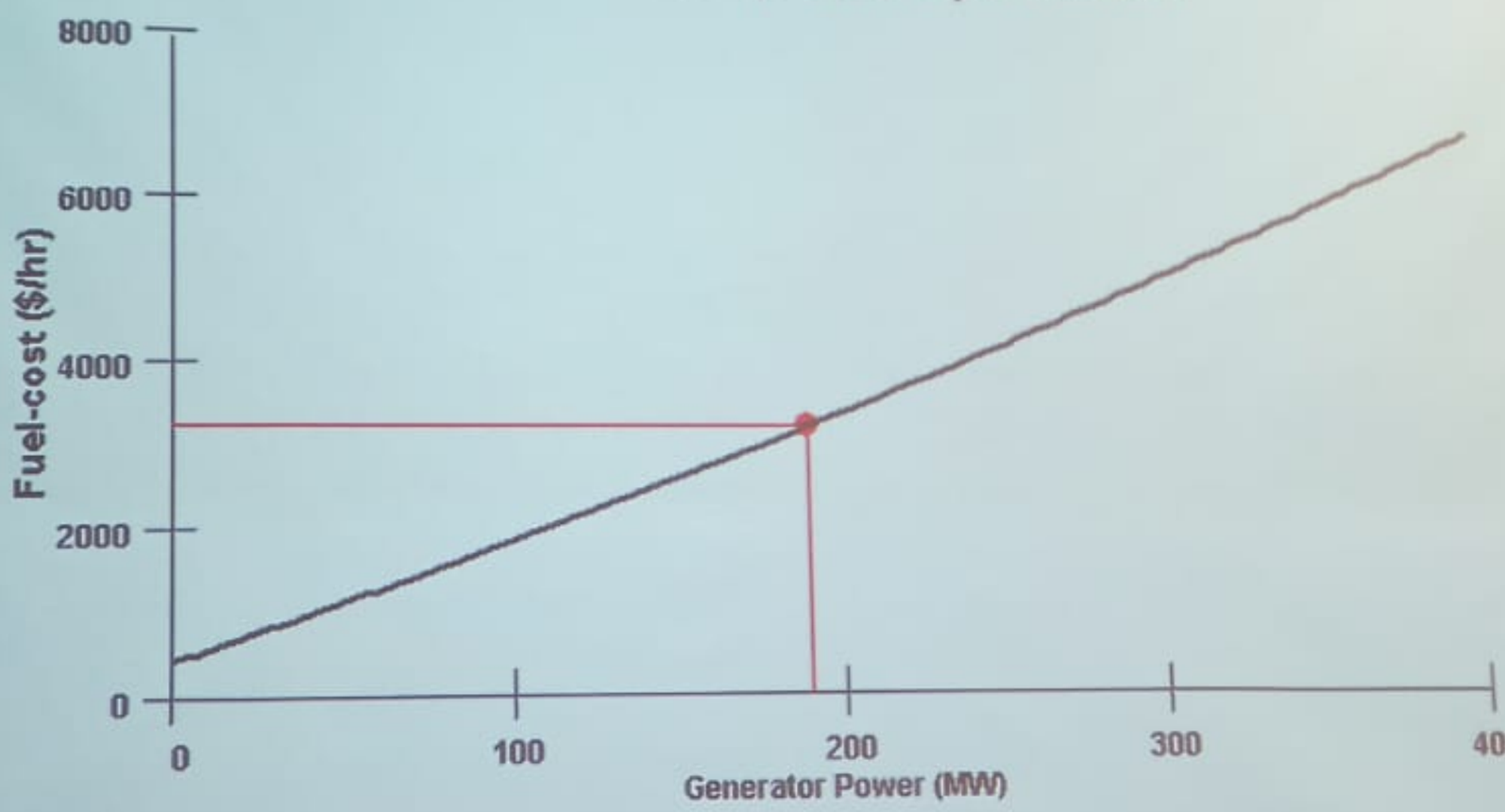
# Fuel-cost Curve

- The fuel-cost curve is the I/O curve multiplied by fuel cost.
- A typical cost for coal is $ 1.70/MBtu.



# Aside: Levelized Cost of Generation, operating costs plus paying off capital.

| Technology | $/MWh (2007 Dollars) (IOU) |
|---|---|
| Advanced Nuclear | 104 |
| Wind – Class 5 | 67 |
| Solar – Photovoltaic | 686 |
| Solar – Concentrating | 434 |
| Solar – Parabolic Trough | 281 |
| Ocean Wave (Pilot) | 838 |
| Small Scale Hydro | 118 |
| Geothermal | 63 |

Keep in mind these numbers involve, assumptions that can drastically affect the value, and that many technology costs are site dependent.

Source: California Energy Commission:
http://energyalmanac.ca.gov/electricity/levelized_costs.html

# Retail Electricity Prices

- There are many fixed and variable costs associated with power systems, which ultimately contribute to determining retail electricity prices.
- The major *variable* operating cost is associated with generation, primarily due to fuel costs:
  - Roughly half of retail costs.

# Incremental Cost Example, cont'd

If $P_{G1} = 250$ MW and $P_{G2} = 150$ MW Then

$$C_1(250) = 1000 + 20 \times 250 + 0.01 \times 250^2 = \$\,6625/\text{hr}$$

$$C_2(150) = 400 + 15 \times 150 + 0.03 \times 150^2 = \$6025/\text{hr}$$

Then

$$IC_1(250) = 20 + 0.02 \times 250 = \$\,25/\text{MWh}$$

$$IC_2(150) = 15 + 0.06 \times 150 = \$\,24/\text{MWh}$$

# Power System Economic Operation

- Fuel cost to generate a MWh can vary widely from technology to technology.
- For some types of units, such as hydro, “fuel” costs are zero but the limit on total available water gives it an implicit value.
- For thermal units it is much easier to characterize costs.
- We will focus on minimizing the variable operating costs (primarily fuel costs) to meet demand.

# Power System Economic Operation

- Power system loads are cyclical.
- Therefore the installed generation capacity is usually much greater than the current load.
- This means that there are typically many ways we could meet the current load.
- Since different states have different mixes of generation, we will consider how generally to minimize the variable operating costs given an arbitrary, specified portfolio of generators.

# Coal Usage Example

- A 500 MW (net) generator is 35% efficient. It is being supplied with coal costing $1.70 per MBtu and with heat content 9000 Btu per pound. What is the coal usage in lbs/hr? What is the cost?

At 35% efficiency required fuel input per hour is

$$\frac{500\ \text{MWh}}{\text{hr}\times 0.35}=\frac{1428\ \text{MWh}}{\text{hr}}\times\frac{1\ \text{MBtu}}{0.29\ \text{MWh}}=\frac{4924\ \text{MBtu}}{\text{hr}}$$

$$\frac{4924\ \text{MBtu}}{\text{hr}}\times\frac{1\ \text{lb}}{0.009\text{MBtu}}=\frac{547{,}111\ \text{lbs}}{\text{hr}}$$

$$\text{Cost}=\frac{4924\ \text{MBtu}}{\text{hr}}\times\frac{\$1.70}{\text{MBtu}}=8370.8\ \$/\text{hr or }\$16.74/\text{MWh}$$

# Power System Economic Operation

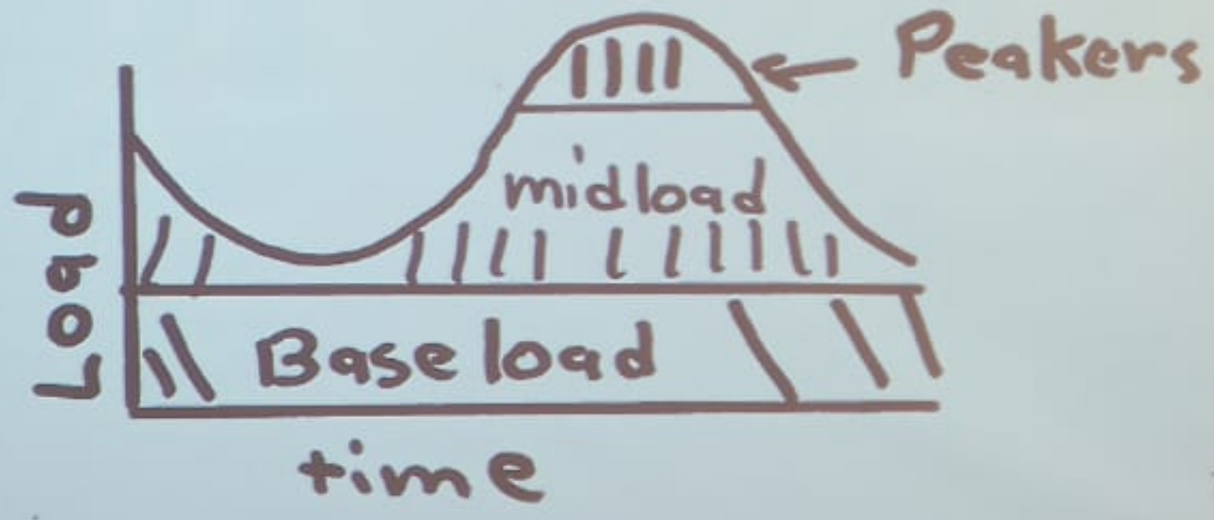
- Different generation technologies vary in the:
  - capital costs necessary to build the generator
  - fuel costs to actually produce electric power
- For example:
  - nuclear and hydro have high capital costs and low operating costs.
  - Natural gas generators have low capital costs, and higher operating costs.

# Thermal versus Hydro Generation

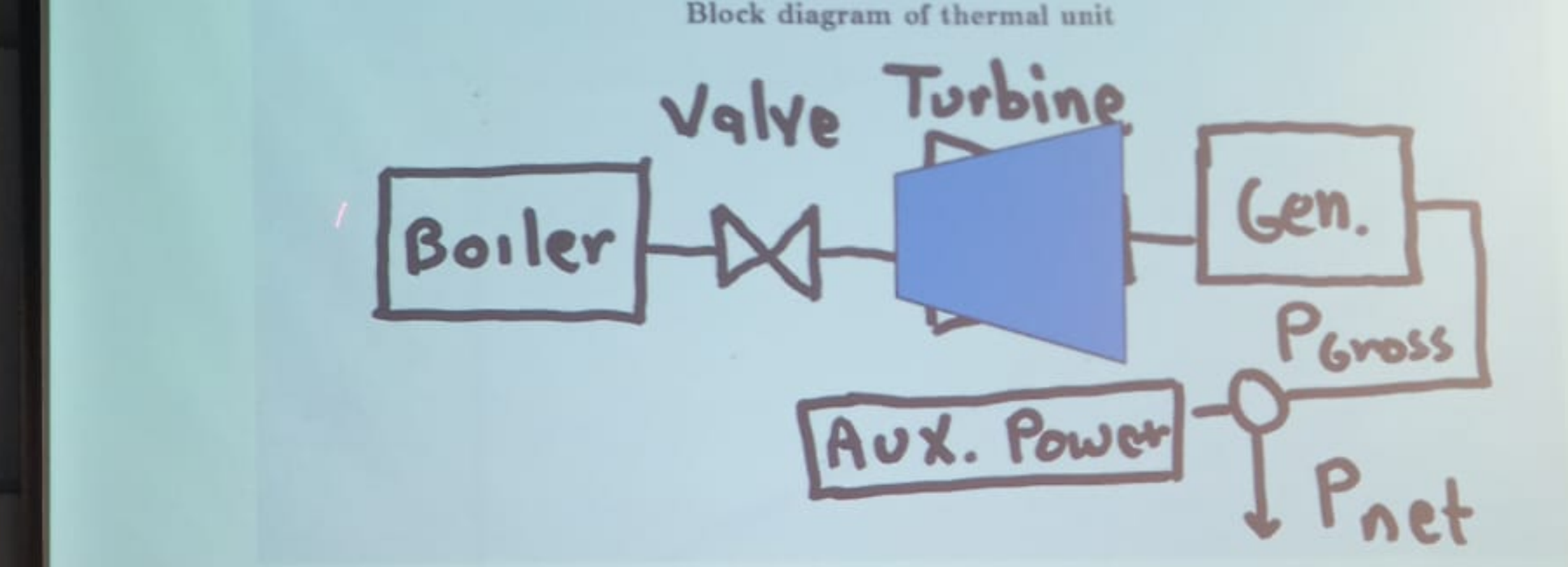
- The two main types of generating units are thermal and hydro, with wind and solar rapidly growing.
- For hydro the fuel (water) is free but there may be many constraints on operation:
  - fixed amounts of water available,
  - reservoir levels must be managed and coordinated,
  - downstream flow rates for fish and navigation.
- Hydro optimization is typically longer term (many months or years).
- We will concentrate on thermal units and some wind, looking at short-term optimization.

# Generator types

- Traditionally utilities have had three broad groups of generators:
  - "Baseload" units: large coal/nuclear; almost always on at max.
  - "Midload," 'intermediate," or "cycling" units: smaller coal or gas that cycle on/off daily or weekly.
  - "Peaker" units: combustion turbines used only for several hours.



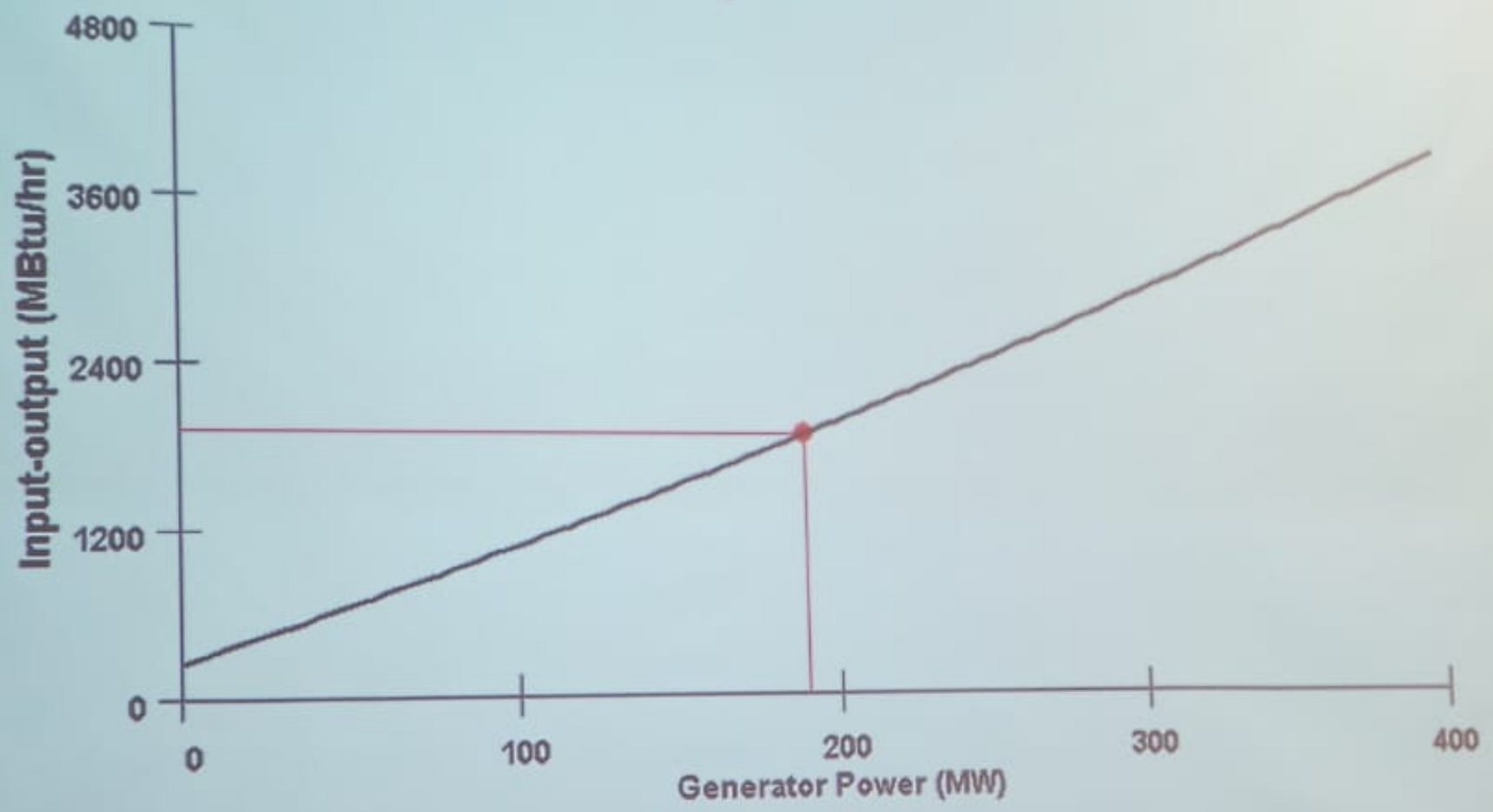
# Block Diagram of Thermal Unit




- To optimize generation costs we need to develop cost relationships between net power out and operating costs.
- Between 2-10% of power is used within the generating plant; this is known as the auxiliary power.

# I/O Curve

- The IO curve plots fuel input (in MBtu/hr) versus net MW output.



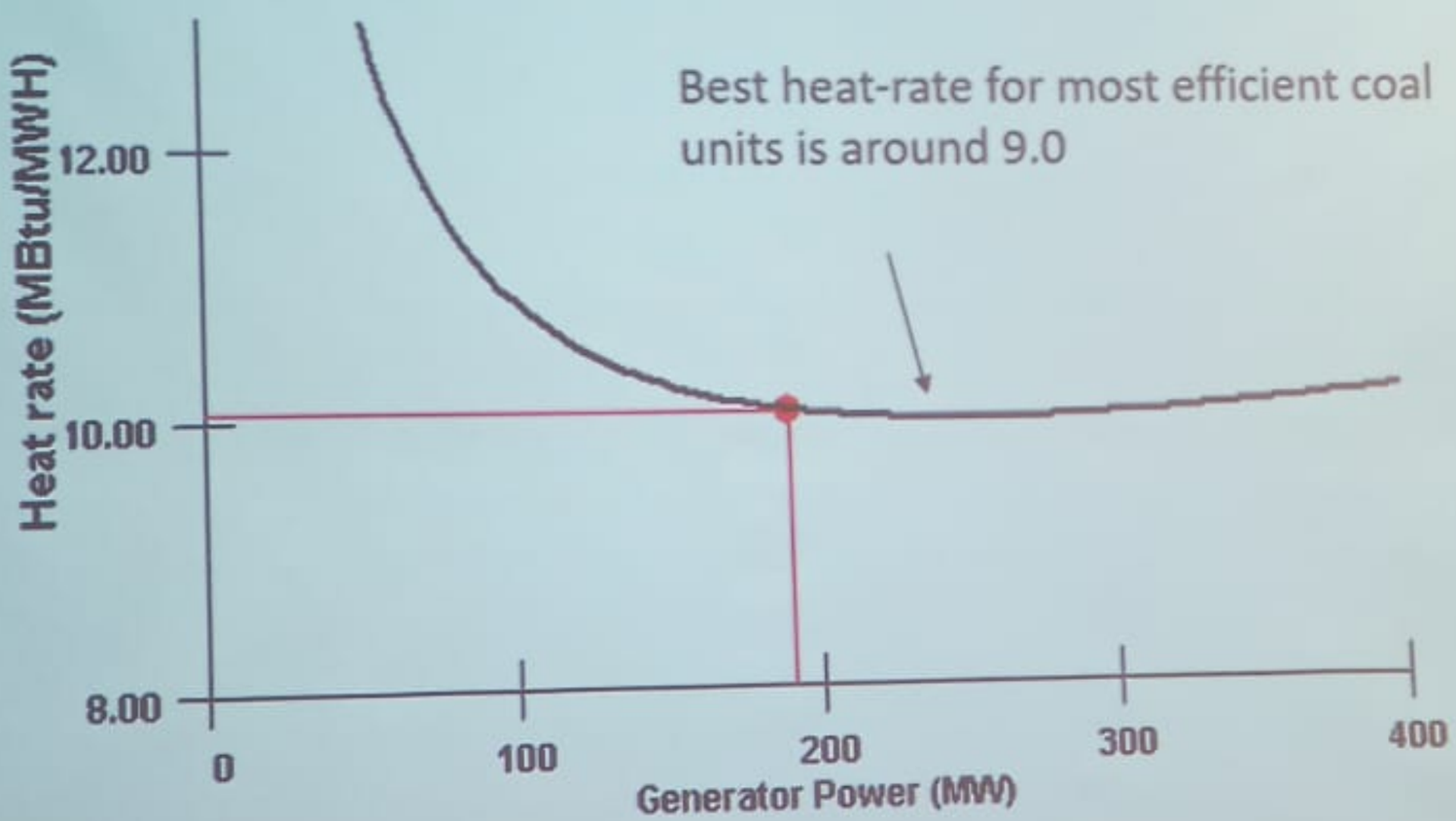
# Thermal generator Cost Curves

- Thermal generator costs are typically represented by one or other of the following four curves
  - input/output (I/O) curve
  - fuel-cost curve
  - heat-rate curve
  - incremental cost curve

- For reference
  - 1 Btu (British thermal unit) = 1054 J
  - 1 MBtu = $1x10^6$ Btu
  - 1 MBtu = 0.29 MWh

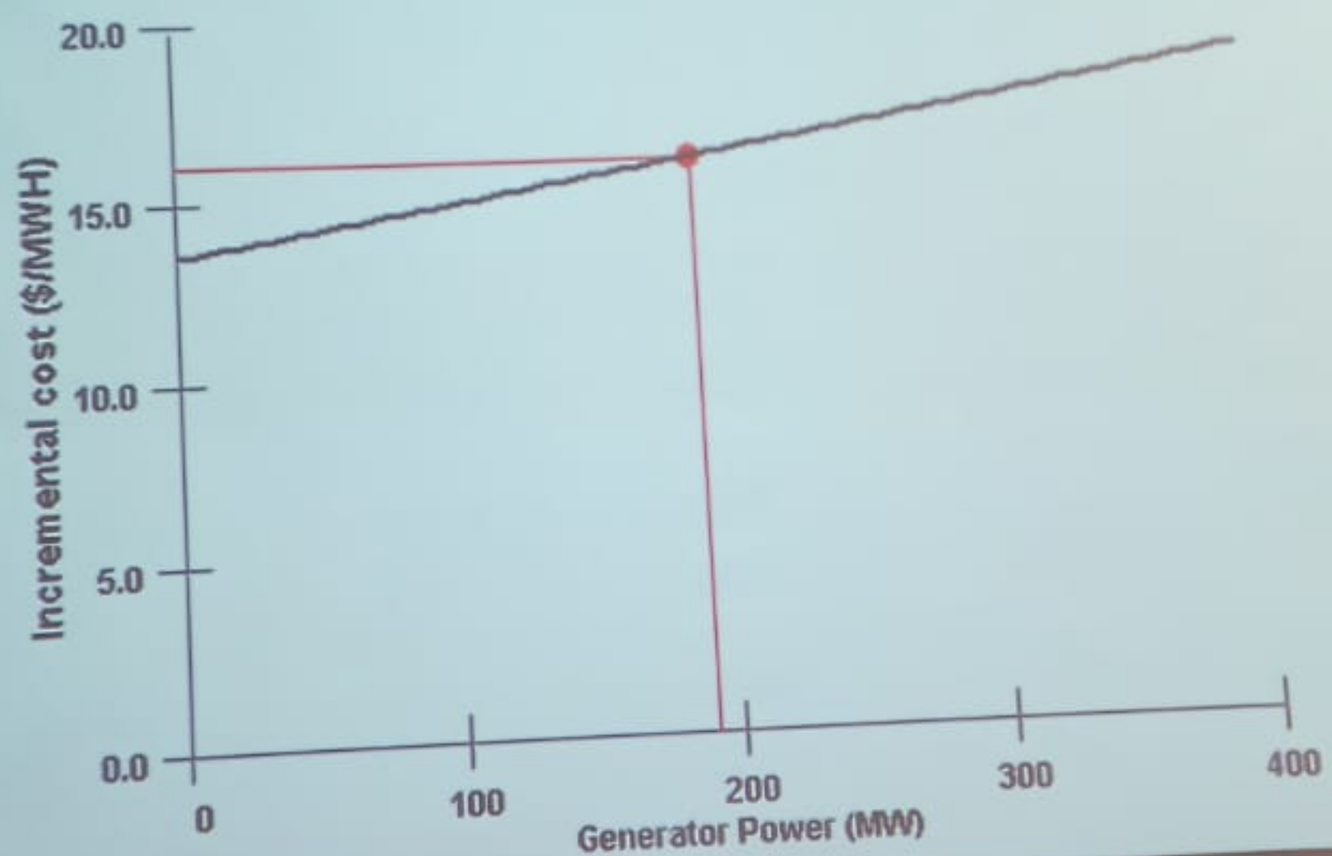
# Heat-rate Curve

- Plots the average number of MBtu/hr of fuel input needed per MW of output.
- Heat-rate curve is the I/O curve divided by MW.



# Incremental (Marginal) cost Curve

- Plots the incremental $/MWh as a function of MW.
- Found by differentiating the cost curve.



# Wasting Coal Example

- Assume a 100W lamp is left on by mistake for 8 hours, and that the electricity is supplied by the previous coal plant and that transmission/distribution losses are 20%. How much coal has he/she wasted?

With 20% losses, a 100W load on for 8 hrs requires 1 kWh of energy. With 35% gen. efficiency this requires

$$\frac{1\ \text{kWh}}{0.35} \times \frac{1\ \text{MWh}}{1000\ \text{kWh}} \times \frac{1\ \text{MBtu}}{0.29\ \text{MWh}} \times \frac{1\ \text{lb}}{0.009\text{MBtu}} = 1.09\ \text{lb}$$

# Mathematical Formulation of Costs

- Generator cost curves are usually not smooth. However the curves can usually be adequately approximated using piece-wise smooth, functions.
- Two approximations predominate:
  - quadratic or cubic functions
  - piecewise linear functions
- We'll assume a quadratic approximation:

$$C_i(P_{Gi}) = \alpha_i + \beta_i P_{Gi} + \gamma_i P_{Gi}^2 \quad \$/\text{hr (fuel-cost)}$$

$$IC_i(P_{Gi}) = \frac{dC_i(P_{Gi})}{dP_{Gi}} = \beta_i + 2\gamma_i P_{Gi} \quad \$/\text{MWh}$$

# Incremental Cost Example

For a two generator system assume

$$C_1(P_{G1}) = 1000 + 20P_{G1} + 0.01P_{G1}^2 \text{ \$/hr}$$

$$C_2(P_{G2}) = 400 + 15P_{G2} + 0.03P_{G2}^2 \text{ \$/hr}$$

Then

$$IC_1(P_{G1}) = \frac{dC_1(P_{G1})}{dP_{G1}} = 20 + 0.02P_{G1} \text{ \$/MWh}$$

$$IC_2(P_{G2}) = \frac{dC_2(P_{G2})}{dP_{G2}} = 15 + 0.06P_{G2} \text{ \$/MWh}$$

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# Economic Dispatch Example

What is economic dispatch for a two generator system $P_D = P_{G1} + P_{G2} = 500$ MW and

$$C_1(P_{G1}) = 1000 + 20P_{G1} + 0.01P_{G1}^2 \quad \$/hr$$

$$C_2(P_{G2}) = 400 + 15P_{G2} + 0.03P_{G2}^2 \quad \$/hr$$

Using the Largrange multiplier method we know

$$\frac{dC_1(P_{G1})}{dP_{G1}} - \lambda = 20 + 0.02P_{G1} - \lambda = 0$$

$$\frac{dC_2(P_{G2})}{dP_{G2}} - \lambda = 15 + 0.06P_{G2} - \lambda = 0$$

$$500 - P_{G1} - P_{G2} = 0$$



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# Economic Dispatch Lagrangian

For the economic dispatch we have a minimization constrained with a single equality constraint

$$L(\mathbf{P}_G,\lambda) = \sum_{i=1}^{m} C_i(P_{Gi}) + \lambda(P_D - \sum_{i=1}^{m} P_{Gi}) \quad \text{(no losses)}$$

The necessary conditions for a minimum are

$$\frac{\partial L(\mathbf{P}_G,\lambda)}{\partial P_{Gi}} = \frac{dC_i(P_{Gi})}{dP_{Gi}} - \lambda = 0 \quad \text{(for i = 1 to m)}$$

$$P_D - \sum_{i=1}^{m} P_{Gi} = 0$$



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# Economic Dispatch Example, cont'd

We therefore need to solve three linear equations

$$20 + 0.02P_{G1} - \lambda = 0$$

$$15 + 0.06P_{G2} - \lambda = 0$$

$$500 - P_{G1} - P_{G2} = 0$$

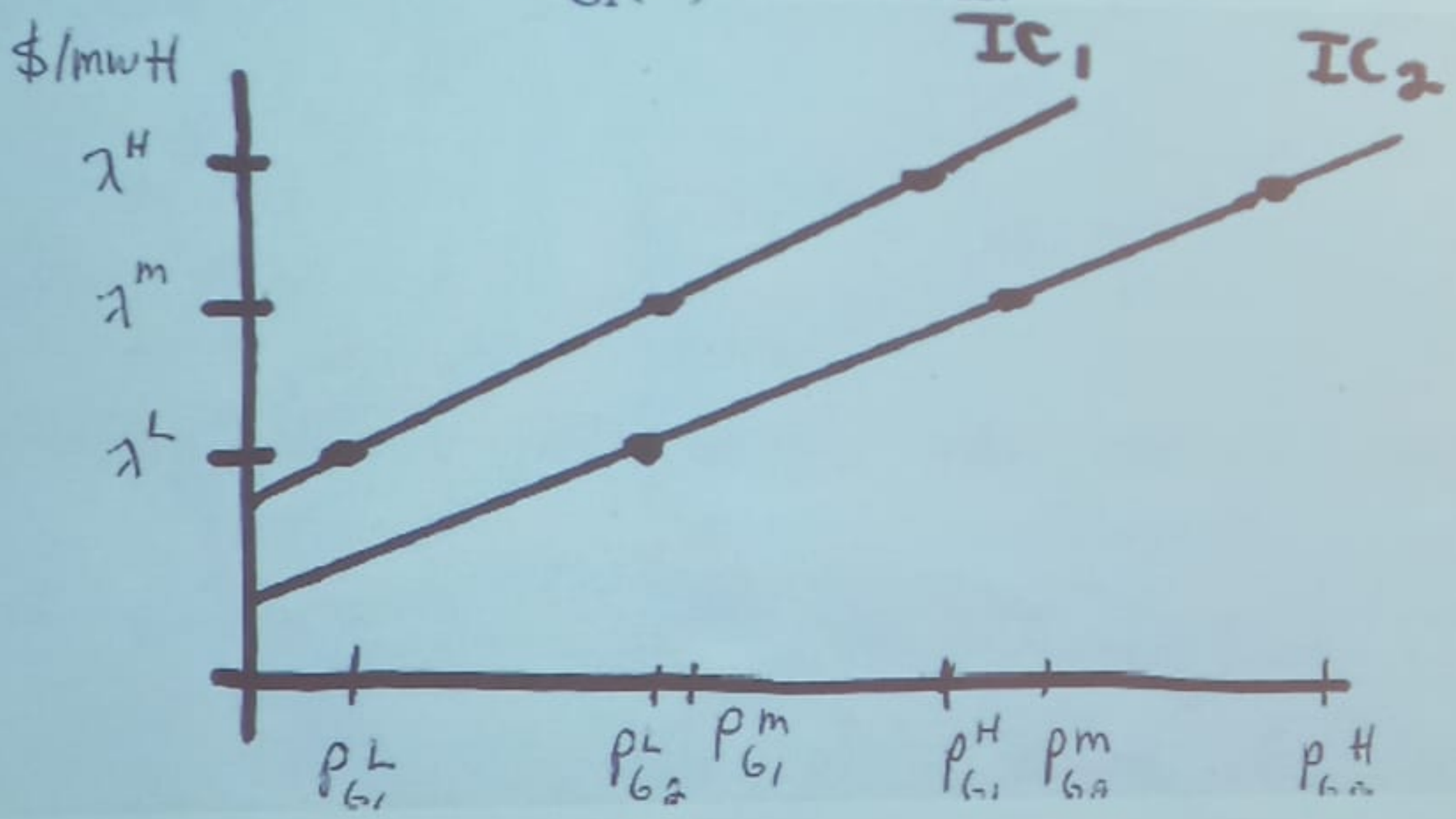
$$\begin{bmatrix} 0.02 & 0 & -1 \\ 0 & 0.06 & -1 \\ -1 & -1 & \end{bmatrix} \begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} -20 \\ -15 \\ -500 \end{bmatrix}$$

$$\begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} 312.5 \text{ MW} \\ 187.5 \text{ MW} \\ 26.2 \text{ \$/MWh} \end{bmatrix}$$

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# Lambda-Iteration: Graphical View

In the graph shown below for each value of lambda there is a unique $P_{Gi}$ for each generator. This relationship is the $P_{Gi}(\lambda)$ function.






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# Lambda-Iteration Example, cont'd

Pick convergence tolerance $\varepsilon = 0.05$ \$/MWh

Then iterate since $\left|\lambda^H - \lambda^L\right| > 0.05$

$$\lambda^M = (\lambda^H + \lambda^L)/2 = 25$$

Then since $\sum_{i=1}^{m} P_{Gi}(25) - 1000 = 280$ we set $\lambda^H = 25$

Since $|25 - 20| > 0.05$

$$\lambda^M = (25 + 20)/2 = 22.5$$

$$\sum_{i=1}^{m} P_{Gi}(22.5) - 1000 = -195 \text{ we set } \lambda^L = 22.5$$



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# Lambda-Iteration Algorithm

Pick $\lambda^{L}$ and $\lambda^{H}$ such that

$$\sum_{i=1}^{m} P_{Gi}(\lambda^{L}) - P_D < 0 \qquad \sum_{i=1}^{m} P_{Gi}(\lambda^{H}) - P_D > 0$$

While $\left|\lambda^{H} - \lambda^{L}\right| > \varepsilon$ Do

$$\lambda^{M} = (\lambda^{H} + \lambda^{L})/2$$

If $\sum_{i=1}^{m} P_{Gi}(\lambda^{M}) - P_D > 0$ Then $\lambda^{H} = \lambda^{M}$

Else $\lambda^{L} = \lambda^{M}$

End While



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# Lambda-Iteration Example

Consider a three generator system with

$$IC_1(P_{G1}) = 15 + 0.02P_{G1} = \lambda \quad \$/\text{MWh}$$

$$IC_2(P_{G2}) = 20 + 0.01P_{G2} = \lambda \quad \$/\text{MWh}$$

$$IC_3(P_{G3}) = 18 + 0.025P_{G3} = \lambda \quad \$/\text{MWh}$$

and with constraint $P_{G1} + P_{G2} + P_{G3} = 1000\,\text{MW}$

Rewriting as a function of $\lambda$, $P_{Gi}(\lambda)$, we have

$$P_{G1}(\lambda) = \frac{\lambda - 15}{0.02} \qquad P_{G2}(\lambda) = \frac{\lambda - 20}{0.01}$$

$$P_{G3}(\lambda) = \frac{\lambda - 18}{0.025}$$

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# Generator MW Limits

- Generators have limits on the minimum and maximum amount of power they can produce
- Often times the minimum limit is not zero. This represents a limit on the generator's operation with the desired fuel type
- Because of varying system economics usually many generators in a system are operated at their maximum MW limits.



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# Lambda-Iteration Gen Limit Example

In the previous three generator example assume the same cost characteristics but also with limits

$0 \le P_{G1} \le 300$ MW $\quad 100 \le P_{G2} \le 500$ MW

$200 \le P_{G3} \le 600$ MW

With limits we get

$$\sum_{i=1}^{m} P_{Gi}(20) - 1000 = P_{G1}(20) + P_{G2}(20) + P_{G3}(20) - 1000$$

$$= 250 + 100 + 200 = -450 \text{ MW (compared to -670MW)}$$

$$\sum_{i=1}^{m} P_{Gi}(30) - 1000 = 300 + 500 + 480 - 1000 = 280 \text{ MW}$$



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# Lambda-Iteration with Gen Limits

In the lambda-iteration method the limits are taken into account when calculating $P_{Gi}(\lambda)$:

if $P_{Gi}(\lambda) > P_{Gi.max}$ then $P_{Gi}(\lambda) = P_{Gi.max}$

if $P_{Gi}(\lambda) < P_{Gi.min}$ then $P_{Gi}(\lambda) = P_{Gi.min}$



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# Lambda-Iteration Example, cont'd

Continue iterating until $\left|\lambda^{H} - \lambda^{L}\right| < 0.05$

The solution value of $\lambda$, $\lambda^{*}$, is 23.53 $/MWh

Once $\lambda^{*}$ is known we can calculate the $P_{Gi}$

$$P_{G1}(23.5) = \frac{23.53 - 15}{0.02} = 426 \text{ MW}$$

$$P_{G2}(23.5) = \frac{23.53 - 20}{0.01} = 353 \text{ MW}$$

$$P_{G3}(23.5) = \frac{23.53 - 18}{0.025} = 221 \text{ MW}$$



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# Lambda-Iteration Limit Example,cont'd

Again we continue iterating until the convergence condition is satisfied. With limits the final solution of $\lambda$, is 24.43 $/MWh (compared to 23.53 $/MWh without limits). The presence of limits will always cause $\lambda$ to either increase or remain the same.

Final solution is

$$P_{G1}(24.43) = 300 \text{ MW}$$
$$P_{G2}(24.43) = 443 \text{ MW}$$
$$P_{G3}(24.43) = 257 \text{ MW}$$



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# Lambda-Iteration Gen Limit Example

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With limits we get

$$\sum_{i=1}^{m} P_{Gi}(20) - 1000 = P_{G1}(20) + P_{G2}(20) + P_{G3}(20) - 1000$$

$$= 250 + 100 + 200 = -450 \text{ MW (compared to -670MW)}$$

$$\sum_{i=1}^{m} P_{Gi}(30) - 1000 = 300 + 500 + 480 - 1000 = 280 \text{ MW}$$



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# Lambda-Iteration Limit Example,cont'd

Again we continue iterating until the convergence condition is satisfied. With limits the final solution of $\lambda$, is 24.43 $/MWh (compared to 23.53 $/MWh without limits). The presence of limits will always cause $\lambda$ to either increase or remain the same.

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# Inclusion of Transmission Losses

- The losses on the transmission system are a function of the generation dispatch. In general, using generators closer to the load results in lower losses
- This impact on losses should be included when doing the economic dispatch
- Losses can be included by slightly rewriting the Lagrangian:

$$L(\mathbf{P}_G, \lambda) = \sum_{i=1}^{m} C_i(P_{Gi}) + \lambda(P_D + P_L(P_G) - \sum_{i=1}^{m} P_{Gi})$$



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# Back of Envelope Values

- Often times incremental costs can be approximated by a constant value:
  - $/MWhr = fuelcost * heatrate + variable O&M
  - Typical heatrate for a coal plant is 10, modern combustion turbine is 10, combined cycle plant is 7 to 8, older combustion turbine 15.
  - Fuel costs ($/MBtu) are quite variable, with current values around 1.5 for coal, 4 for natural gas, 0.5 for nuclear, probably 10 for fuel oil.
  - Hydro, solar and wind costs tend to be quite low, but for this sources the fuel is free but limited



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# Impact of Transmission Losses

This small change then impacts the necessary conditions for an optimal economic dispatch

$$L(\mathbf{P}_G,\lambda) = \sum_{i=1}^{m} C_i(P_{Gi}) + \lambda(P_D + P_L(P_G) - \sum_{i=1}^{m} P_{Gi})$$

The necessary conditions for a minimum are now

$$\frac{\partial L(\mathbf{P}_G,\lambda)}{\partial P_{Gi}} = \frac{dC_i(P_{Gi})}{dP_{Gi}} - \lambda(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}) = 0$$

$$P_D + P_L(P_G) - \sum_{i=1}^{m} P_{Gi} = 0$$



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# Calculation of Penalty Factors

Unfortunately, the analytic calculation of $L_i$ is somewhat involved. The problem is a small change in the generation at $P_{Gi}$ impacts the flows and hence the losses throughout the entire system. However, using a power flow you can approximate this function by making a small change to $P_{Gi}$ and then seeing how the losses change:

$$\frac{\partial P_L(P_G)}{\partial P_{Gi}} \approx \frac{\Delta P_L(P_G)}{\Delta P_{Gi}}$$

$$L_i \approx \frac{1}{1-\frac{\Delta P_L(P_G)}{\Delta P_{Gi}}}$$



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# Impact of Transmission Losses

The condition for optimal dispatch with losses is then

$$L_1 IC_1(P_{G1}) = L_2 IC_2(P_{G2}) = L_m IC_m(P_{Gm}) = \lambda$$

Since $L_i = \dfrac{1}{\left(1 - \dfrac{\partial P_L(P_G)}{\partial P_{Gi}}\right)}$ if increasing $P_{Gi}$ increases

the losses then $\dfrac{\partial P_L(P_G)}{\partial P_{Gi}} > 0 \rightarrow L_i > 1.0$

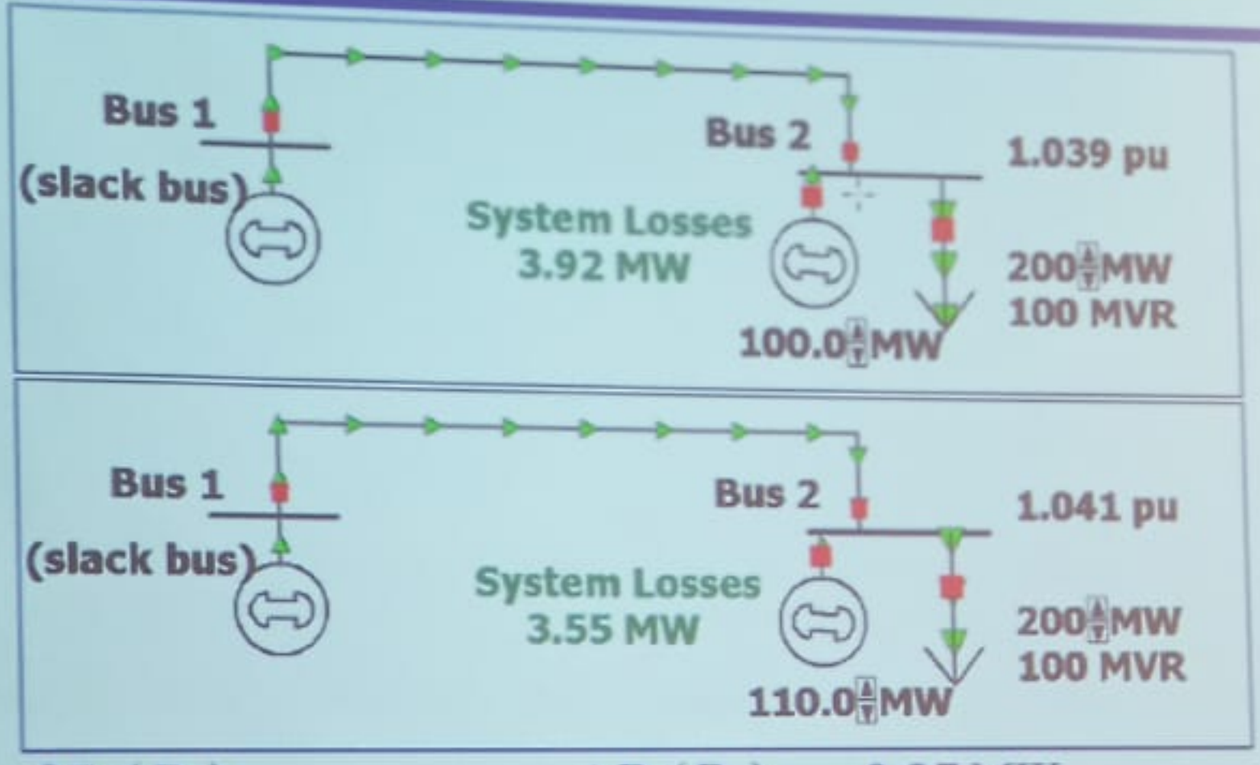
This makes generator i appear to be more expensive (i.e., it is penalized). Likewise $L_i < 1.0$ makes a generator appear less expensive.



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# Two Bus Penalty Factor Example




$$\frac{\partial P_L(P_G)}{\partial P_{G2}} = -0.0387 \qquad \frac{\Delta P_L(P_G)}{\Delta P_{Gi}} = \frac{-0.37MW}{10MW} = -0.037$$

$$L_2 = 0.9627 \qquad L_2 \approx 0.9643$$



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# Economic Dispatch - Summary

- Economic dispatch determines the best way to minimize the current generator operating costs
- The lambda-iteration method is a good approach for solving the economic dispatch problem
  - generator limits are easily handled
  - penalty factors are used to consider the impact of losses
- Economic dispatch is not concerned with determining which units to turn on/off (this is the unit commitment problem)
- Economic dispatch ignores the transmission system limitations



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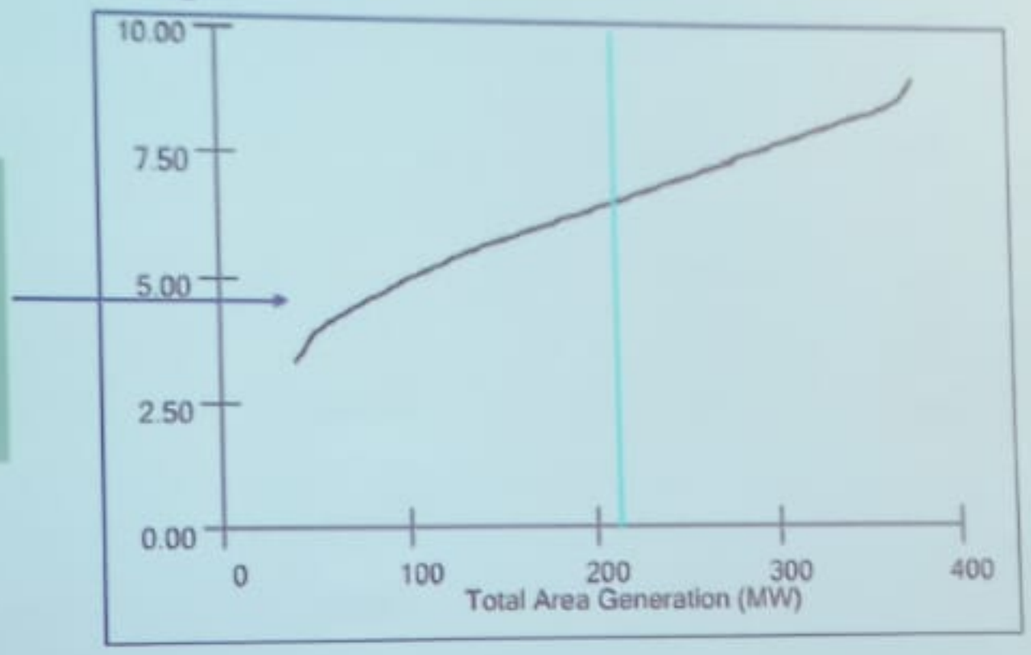
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# Area Supply Curve

- The area supply curve shows the cost to produce the next MW of electricity, assuming area is economically dispatched

Supply curve for thirty bus system



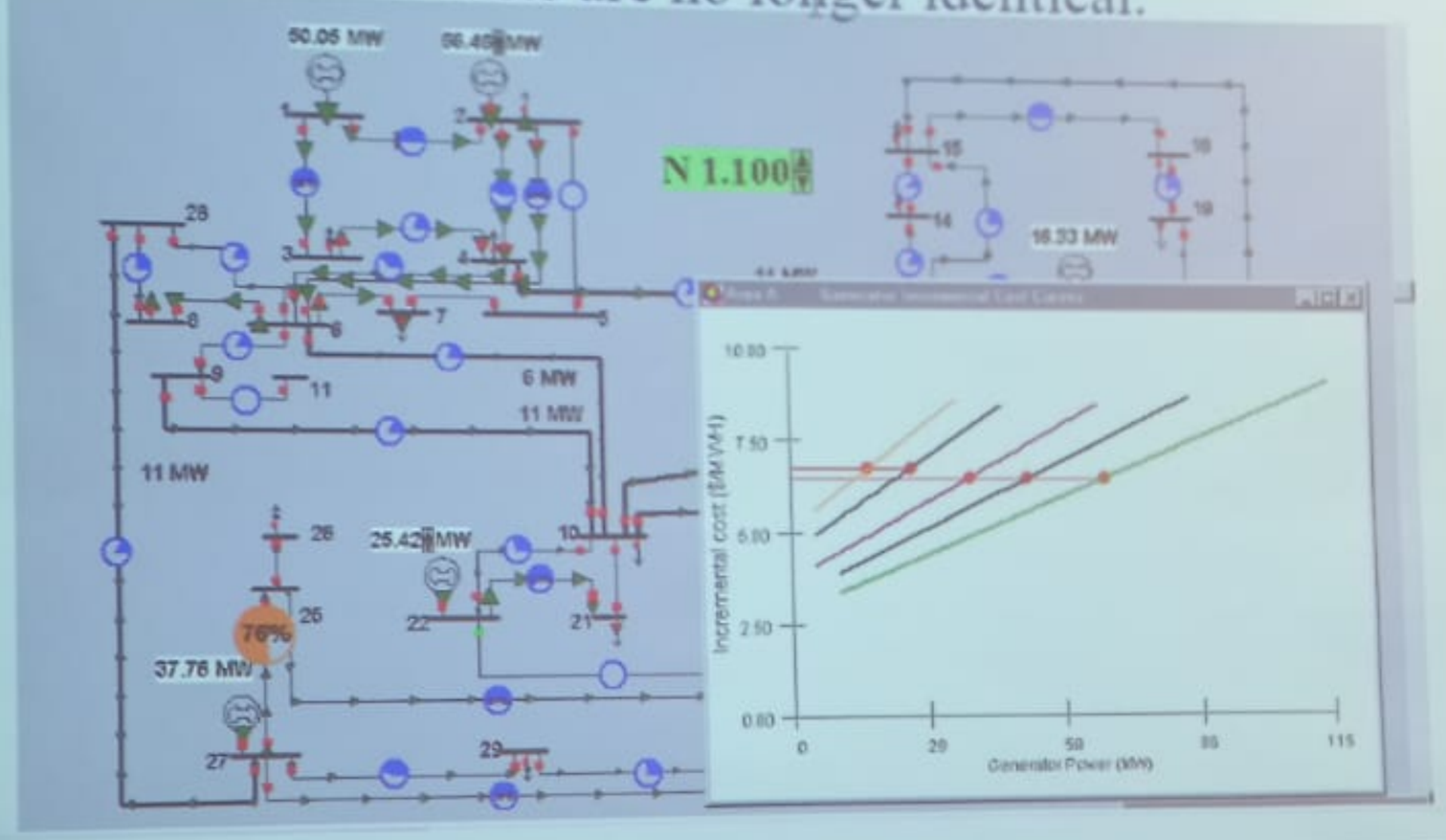



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# Thirty Bus ED Example, cont

- Because of the penalty factors the generator incremental costs are no longer identical.






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- Example
  - consider the cost for operating three generation units

| | | |
|---|---|---|
| Unit 1: | $F_1(P_1) = 561 + 7.92\,P_1 + 0.001562\,P_1^2$ | $150 \le P_1 \le 600$ |
| Unit 2: | $F_2(P_2) = 310 + 7.85\,P_2 + 0.00194\,P_2^2$ | $100 \le P_2 \le 400$ |
| Unit 3: | $F_3(P_3) = 93.6 + 9.56\,P_3 + 0.005784\,P_3^2$ | $50 \le P_3 \le 200$ |

  - what combination of units is best to supply a 550 MW load?

| Unit 1 | Unit 2 | Unit 3 | Max Gen | Min Gen | $P_1$ | $P_2$ | $P_3$ | $F_1$ | $F_2$ | $F_3$ | $F_{total}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Off | Off | Off | 0 | 0 | Infeasible | | | | | | |
| Off | Off | On | 200 | 50 | Infeasible | | | | | | |
| Off | On | Off | 400 | 100 | Infeasible | | | | | | |
| Off | On | On | 600 | 150 | 0 | 400 | 150 | 0 | 3760 | 1658 | 5418 |
| On | Off | Off | 600 | 150 | 550 | 0 | 0 | 5389 | 0 | 0 | 5389 |
| On | Off | On | 800 | 200 | 500 | 0 | 50 | 4911 | 0 | 586 | 5497 |
| On | On | Off | 1000 | 250 | 295 | 255 | 0 | 3030 | 2440 | 0 | 5471 |
| On | On | On | 1200 | 300 | 267 | 233 | 50 | 2787 | 2244 | 586 | 5617 |

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Araçlar İmzala Yorum

# Economic Dispatch - Summary

- Economic dispatch determines the best way to minimize the current generator operating costs
- The lambda-iteration method is a good approach for solving the economic dispatch problem
  - generator limits are easily handled
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